

Cooper and that he was out of town the day Cooper was killed. Williams also states that he saw both Smith and Jones with Cooper the day of the killing and that either Smith or Jones must have killed him. Can you determine who the murderer was if

- a) one of the three men is guilty, the two innocent men are telling the truth, but the statements of the guilty man may or may not be true?
 b) innocent men do not lie?

Soln: Let p : Smith kills Cooper q : Jones kills Cooper r : Williams kills Cooper s : Cooper was a friend of Jones t : Williams likes Smith
 u : Jones ~~was out of town the day Cooper was killed~~ knows Cooper v : Jones was out of town the day Cooper was killed w : Smith was with Cooper on the day of the killing x : Jones was with Cooper on the day of the killing.

Smith says: $\neg p$ and $s \wedge \neg t$ Jones says: $\neg u \wedge v$ Williams says: $(w \wedge x) \wedge (p \vee q)$
 Suppose, Smith is innocent. $\therefore s \wedge \neg t$ is True. By defn. of AND, s is True and $\neg t$ is True.

Its self evident that $[s \Rightarrow u]$ must always be True. By defn. of conditional statement, u must be True. $\therefore$ By defn. of negation, $\neg u$ is False. By defn. of AND, $\neg u \wedge v$ is False. $\therefore$ Jones lies. Jones is guilty.

By problem constraints, William is innocent. $\therefore (w \wedge x) \wedge (p \vee q)$ must be True. By defn. of AND, each of $w, x, (p \vee q)$ is True. This is consistent.

$\therefore$ If Smith is innocent, then Jones must be guilty

Suppose, Jones is innocent. $\therefore \neg u \wedge v$ is True. By defn. of AND, $\neg u$ is True and v is True. Its easy to see $x = \neg v$. By defn. of negation, x is False. By defn. of AND, $(w \wedge x)$ must be False. By defn. of AND, $(w \wedge x) \wedge (p \vee q)$ is False.

$\therefore$ William lies, and William is guilty. By defn. of contrapositive and previous proof, $[s \Rightarrow u]$ is equivalent to $[u \Rightarrow \neg s]$. $\therefore s \Rightarrow u$ is True so is $u \Rightarrow \neg s$

By defn. of conditional, $\neg s$ is True. By defn. of negation, s is False. By defn. of AND, $s \wedge \neg t$ is False.